

Chapter 1

Introduction

Computer Networking:- Computer network is a types of network in which all computers are interconnected with high intelligence peripheral device such as switch, router and telephone line for communication purpose and sharing resources like printers, files and electronic devices.

Advantages and Disadvantages of computer network:

Advantages:

- We can easily share file and printer so it saves our time and money.
- We don't need big storage drive for saving data.
- Every authorized user can easily access data.
- Data is easy to backup as all the data is stored on the file server.

Disadvantages:

- Data managing in network could require person who had many experience and higher education.
- In the internet, people can easily attached virus in data and share the data in intentional.
- The network requires high secure program or system.

Application of Networking:- Some of the network applications in different fields are the following:

- Marketing and sales
- Bank
- University
- Financial services
- Manufacturing
- Electronic messaging
- Directory services
- Information services
- Electronic data interchange (EDI)

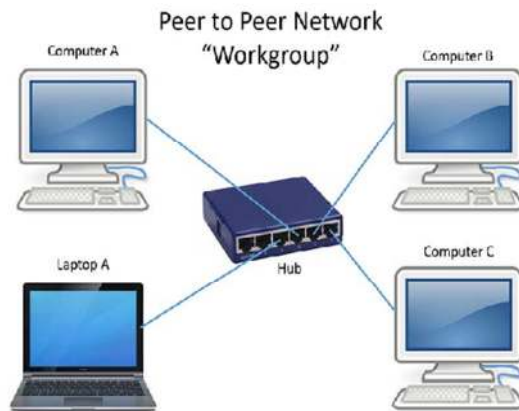
Network Architecture:-

Network architecture is the design of a computer network in which all computers are organized physically and logically in network.

There are two types of network architecture.

1. Peer to peer network
2. client/server network

1. peer to peer network:- Peer to peer network is a types of network in which all computers works as server and client. In this model all computers have equal responsibilities.



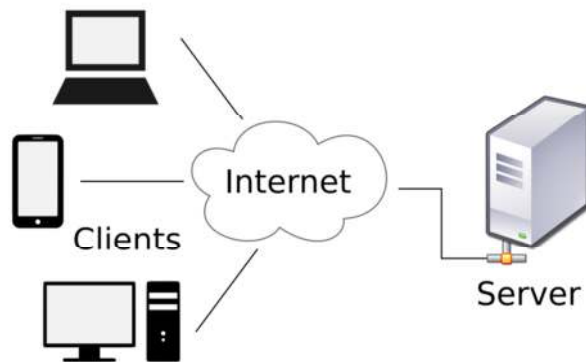
Advantages:-

- Easy to implement and manage.
- All computers are independent with each other.
- If one computer (peer) fails, it will not affect the working of others.
- There is no need for any professional expert such kind of networks.
- The cost of implementation of such networks is very less.

Disadvantages:

- Decentralize management.
- No data backup options are available in peer-to-peer networks.
- These kinds of networks are not secure.

2. client/server network:- client/server network is a types of network in which one dedicated computer exists as server and remaining all computers are clients.



Advantages:

- All computers are managed centrally by server
- All data is saved on the central area.
- Data can be easily backed in such networks.
- Servers can serve multiple clients at a time.
- Data and computers are handled only by authorized users.

Disadvantage:

- Dependency is more on a centralized server.
- A network administrator is required for handle entire network.
- If the server fails, all clients will be affected.
- High cost.

Differentiate Between client and server

Client	Server
There are no dedicated server exists.	There are one dedicated server exists.
Client computer accept the services.	Server computer provide the services.
In this model, all computers are decentralized.	In this model, all the computers are managed centrally.
It is easy to setup and handle	It is difficult to setup and manage.
It is less secure.	It is more secure.
There is no need network administrator for operate and manage.	There is required network administrator for operate and manage.
If one computer fail, no others computer	If one server computer fails, the entire client

effected.	computer effected.
Example: windows xp/windows 7/windows 10/11	Example: windows server 2003/2008/2012/2016/2019/2022/linux/unix etc.

Ethernet cross-over cable

In this cable, transmitting pins of one side connect with the receiving pins of the other side.

The wire at pin 1 on one end of the cable connects to pin 3 at the other end of the cable. The wire at pin 2 connects to pin 6 on the other end of the cable. Remaining wires connect in the same positions at both ends.

The following table lists the wire positions of the cross-over cable on both sides.

Side A	Side B
Green White	Orange White
Green	Orange
Orange White	Green White
Blue	Blue
Blue White	Blue White
Orange	Green
Brown White	Brown White
Brown	Brown

The cross-over cable is used to connect the following devices.

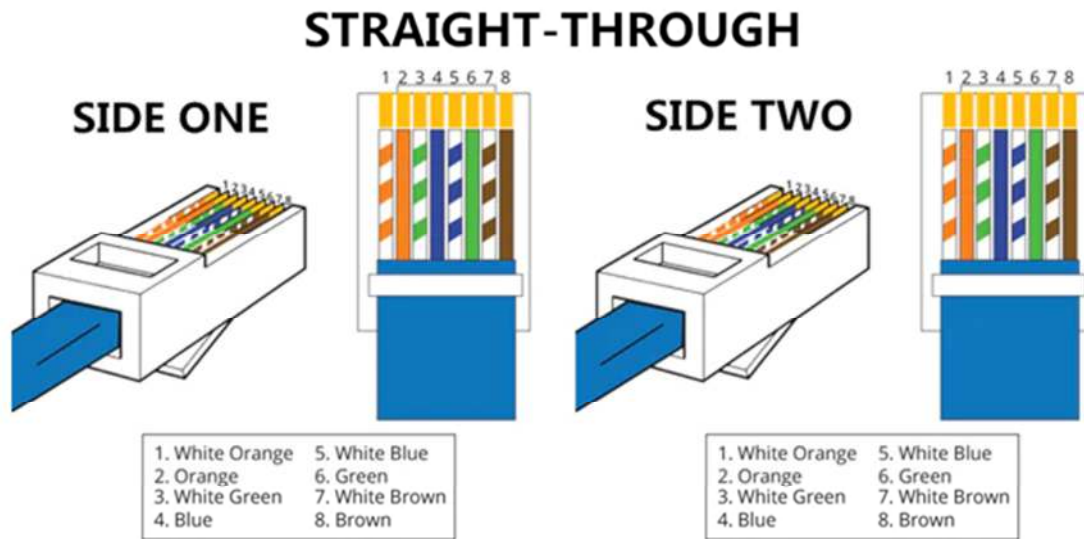
- Two computers
- Two hubs
- A hub to a switch
- A cable modem to a router
- Two router interfaces

Straight Cabling

Straight-through cable is a type of twisted pair copper wire cable for local area network (LAN) use for which the RJ-45 connectors at each end have the same pinout (i.e., arrangement of conductors).

- PC to Switch Straight
- Router to hub Straight
- Computer to computer straight

LED Rating Forward Voltage: 2.2 Volts Typical.



Types of network:- There are several different types of computer networks. Computer networks can be characterized by their size as well as their purpose. According to size, there are following types of computer network.

1. PAN
2. LAN
3. CAN
4. MAN
5. WAN

1. PAN (personal)

A personal area network (PAN) connects electronic devices close to the user, such as a wireless mouse, a keyboard, and a computer.

- It is used for data transmission among devices such as computer, mobile etc.
- A personal area network (PAN) connects electronic devices within a user's immediate area.
- The size of a PAN ranges from a few centimeters to a few meters.

Advantages:

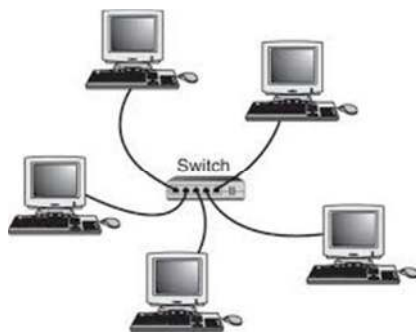
- Allows for easy communication between personal devices in close proximity.
- Can be set up easily and quickly.

Disadvantages:

- Limited coverage area.
- May not be suitable for large-scale data transfer or communication.

2. **CAN (Campus Area Network)** is used mainly within a site which connects different buildings and departments.

3. **Local area network:** - A local area network is a network in which all computers are confined within a single room or a single building. A local area network may be wired, wireless or may be both.



Advantages:

- Provides fast data transfer rates and high-speed communication.
- Easy to set up and manage.
- Low cost setup
- Provides increased security and fault tolerance compared to WANs.
- Troubleshooting of LAN networks is easy.

Disadvantages:

- Limited geographical coverage.
- Due to the sharing of computer resources among the devices, sometimes the operation speed of the network may slow down.

4. **Metropolitan Area Network:** - A metropolitan area network is larger than LAN and smaller than WAN. In MAN various local area networks are connected with each other within city. The most widely used protocols in MAN are RS-232, Frame Relay, ATM, ISDN, OC-3 etc.



Advantages:

- Wider Coverage
- Higher Security
- Scalability
- Improved Communication
- Faster Data Transfer Speed
- Simultaneous Two-Way Data Transmission
- Internet Sharing
- Cost Savings in WAN Setup
- Centralized Management
- Strong Backbone Network for Large Systems
- Easy to Convert from LAN to MAN

Disadvantages:

- Required wires
- Security Risks
- Higher Costs
- Difficult to manage
- Limited Connectivity
- Need for Skilled Technicians

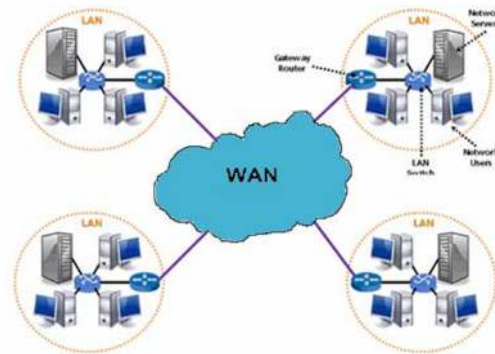
Metropolitan Area Network Example

There are various examples of a MAN network that we can observe daily. Some of those are:

- Cable TV network
- Telephone networks
- DSL line
- The IUB network

- IEEE 802.16
- WiMAX

3. Wide area network:- A wide area network (WAN) cover large geographical area that may comprise a country, a continent, or even the whole world. A wide area network is possible by using router. Router is the process of transforming data from one network to another network.



Advantages:

- **Large Network Range:** It cover a large geographical area from one country to another countries.
- **Centralized data:** It is used to connect different branch offices and sharing data. Thus, we don't need to buy email servers, files server and back up servers, etc.
- **Get updated files and data:** It provides an ideal platform for companies who need a live server for their employees to exchange updated files within seconds.
- **High bandwidth:** It offers high bandwidth than a normal broadband connection. Thus, it can increase the productivity of our company by offering uninterrupted data transfer and communication.
- **Workload Distribution:** It helps distribute your workload to other locations. You can hire employees in different countries and assign them to work from your office.

Disadvantages:

- **Security issue:** A WAN network has more security issues as compared to LAN and MAN network as all the technologies are combined together that creates the security problem.
- **Needs Firewall & antivirus software:** The data is transferred on the internet which can be changed or hacked by the hackers, so the firewall needs to be used. Some people can inject the virus in our system so antivirus is needed to protect from such a virus.
- **High Setup cost:** An installation cost of the WAN network is high as it involves the purchasing of routers, switches.

- **Troubleshooting problems:** It covers a large area so fixing the problem is difficult.

Protocols and Standards

Protocol:- Protocol is a set of rules for communicating between two or more devices. The protocol defines a common set of rules which are used by computers on the network that communicate between hardware and software entities. One of the most popular protocols for LANs is the Ethernet. Another popular LAN protocol for PCs is the token-ring network. The key elements of a protocol are

1. Syntax
2. Semantics
3. Timing

1. Syntax:-

- Structure or format of the data.
- Indicates how to read the bits

2. Semantic:-

- Interpreted the meaning of bits
- How a particular pattern is interpreted and what action to be taken.
(Example, does an address identify the route to be taken or the final destination of the message?)

3. Timing:-

- When data should be sent and how fast they can be sent.
(The term timing refers to two characteristics: when data should be sent and how fast they can be sent).

Standards:

Standards provide guidelines to manufactures, vendors, government agencies, and other service providers.

Data communication standards fall into two categories:

1. de facto
2. de jure

1. de facto :-

Standards that have not been approved by an organized body but have been adopted as standards through widespread use are de facto standards.

2. de jure:-

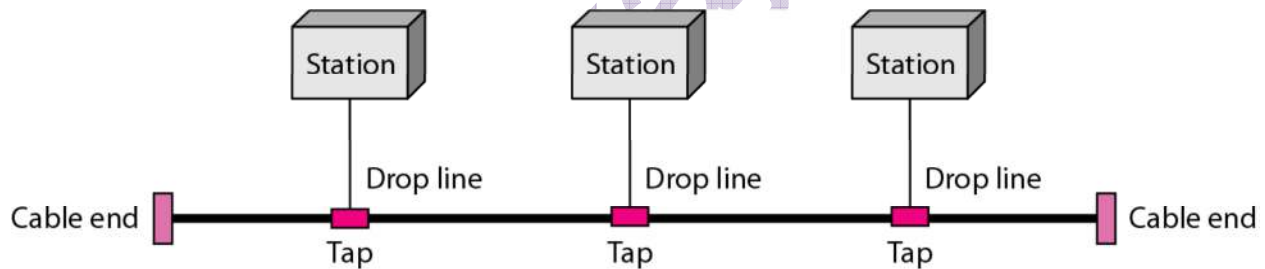
Those standards by law or by regulation. These are the standards recognized officially by an Organization.

Some Standard Creation committees are :

1. International Organization of Standardization(ISO)
2. American National Standards Institute(ANSI)
3. Electronic Industries Association(EIA)
4. Institute of Electrical and Electronics Engineers(IEEE)

Network Topologies:- The arrangement of physical connectivity of entire network is called topology. The different topologies are given below.

1. Bus topology:- A topology in which all computers are connected in a single cable or a single line is called bus topology or it is also called linear topology.



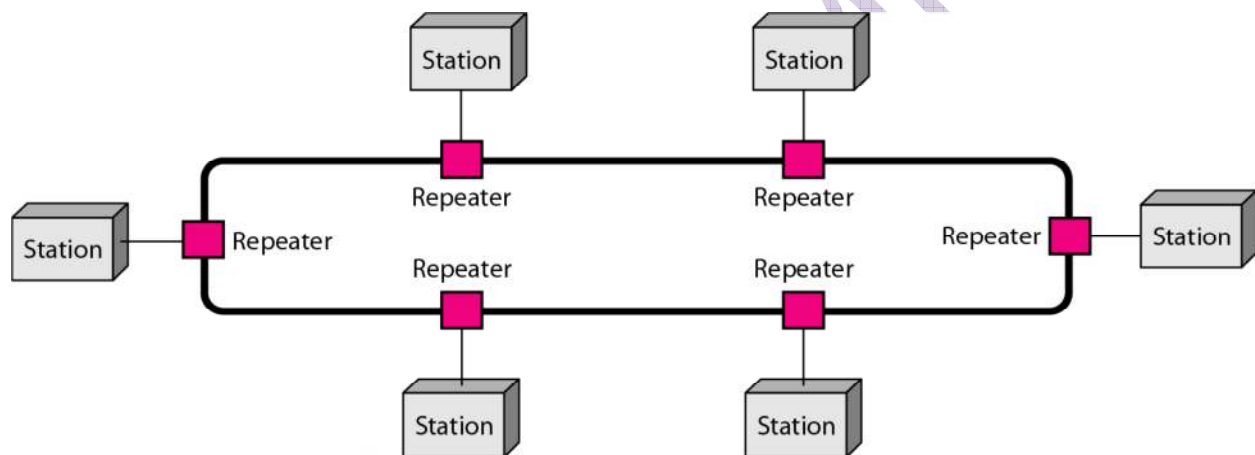
Advantages:-

- It is easy to set-up and extend bus network.
- Cable length required for this topology is the least compared to other networks.
- Bus topology very cheap.
- Linear Bus network is mostly used in small networks.

Disadvantages:-

- It can be hard to troubleshoot individual device issues.
- Bus topology is not great for large networks.
- Terminators are required for both ends of the main cable.
- If a main cable is damaged, the network fails or splits into two.

2. Ring Topology:- A topology in which all computers are connected in a circular ring is called ring topology. There are no terminated ends. A ring topology connects one host to the next and the last host to the first. The signal travels around the loop in one direction and pass through each computer.



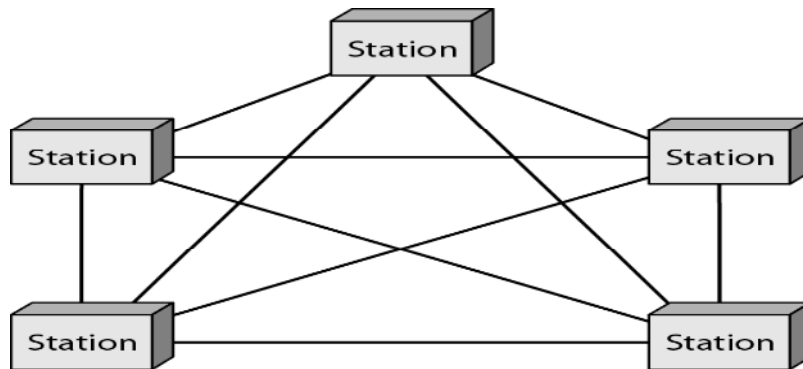
Advantages:

- All data flows in one direction, reducing the chance of packet collisions.
- A network server is not needed to control network connectivity between each workstation.
- Data can transfer between workstations at high speeds.

Disadvantages:

- Each packet of data must pass through all the computers between source and destination. This makes it slower than Star topology.
- If one workstation or port goes down, the entire network gets affected.

3. Mesh Topology:- A topology in which all computers are connected to each other.



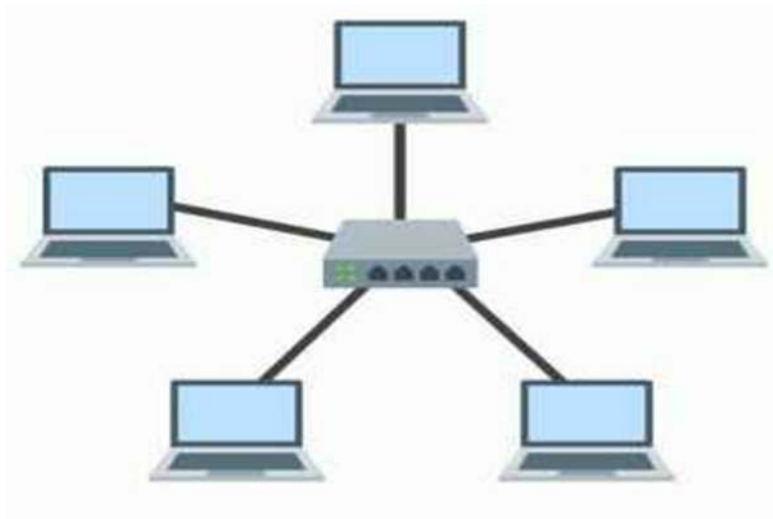
Advantages:-

- Can handle high amounts of traffic, because multiple devices can transmit data simultaneously.
- A failure of one device does not affect another device.
- Adding additional devices does not disrupt data transmission between other devices.

Disadvantages:-

- It is very expensive.
- Building and maintaining the topology is difficult and time consuming.

4. Star Topology:- A topology in which all computers are connected centrally by using switch. Signals are transmitted from the sending computer through the hub or switch to all computers on the network. This topology originated in the early days of computing and it is very popular. It is now a common topology in microcomputer networking.



Advantages:-

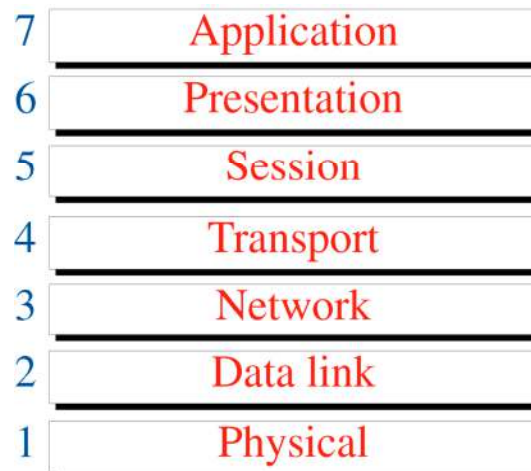
- Centralized management of the network, through the use of the central computer, hub, or switch.
- Easy to add another computer to the network.
- If one computer on the network fails, the rest of the network continues to function normally.

Disadvantages:-

- Used more cable.
- The use of hub, a router or a switch as central device increases the overall cost of the network.
- If the central computer, hub, or switch fails, the entire network goes down and all computers are disconnected from the network

OSI Model:- The Open Systems Interconnection (OSI) model was developed by the International Organization for Standardization (ISO), and formalized in 1984. It provided the first framework governing how information should be sent across a network.

OSI stands for open system interconnection model it tells us how data travel in a network. OSI model has 7 layers.



1. Physical Layer:- The first layer of OSI model is called physical layer. The physical layer describes electrical or optical signals. All the physical connectivity of network takes place such as cable RJ45, clamping. It converts data into bits and forward it to the physical layer to the receiving sides.

Devices used :

- Hub
- Repeater

2. Data-Link layer:-The second layer of the OSI model is called data-link layer. The data link layer establishes a connection between two physically connected devices on a network. It is responsible for node-to-node delivery of data. The Data-Link layer consists of two sub layers:

- Logical Link Control (LLC) sub layer.
- Media Access Control (MAC) sub layer.

Logical Link Layer (LLC) : It is responsible for transferring the packets to the Network layer of the receiver that is receiving.

Medium Access Control (MAC): establishes and maintains links between communicating devices.

Protocol used under this layer are: ppp(point to point protocol), PPTP(point to point tunneling protocol), ARP(address resolution protocol) and RARP(Reverse address resolution protocol).

Devices:

- Switch

3. Network Layer:- The 3rd layer of OSI Network mode is called Network layer. The network layer is responsible for the source-to-destination delivery of a packet. The network layer has two main functions.

1. Logical addressing –Logical address provides a unique address that identifies computers on internet.
2. Routing – Router determines the best path to send and receive data from source to destination. Protocol used under the layer are:- ip, RIP, IGRP, EIGRP, OSPF etc.

Devices:

- Router

4. Transport layer:- the 4th layer of OSI network model is called transport layer. It is responsible for process-to-process delivery of the entire message. Transport layer uses two protocols.

1. TCP/IP :- It established the connection between hosts.
2. UDP:- It is unreliable protocol.

The transport layer has the following functionality.

1. Segmentation.
2. Flow control
3. Error control.
4. Sequencing

5. Session Layer:- the 5th layer of OSI model is called session layer. The Session layer is responsible for dialog control, establishing the connection, manage the connection and terminate the connection.

Sessions layer also controls whether to send the data in simplex, half duplex and full duplex.

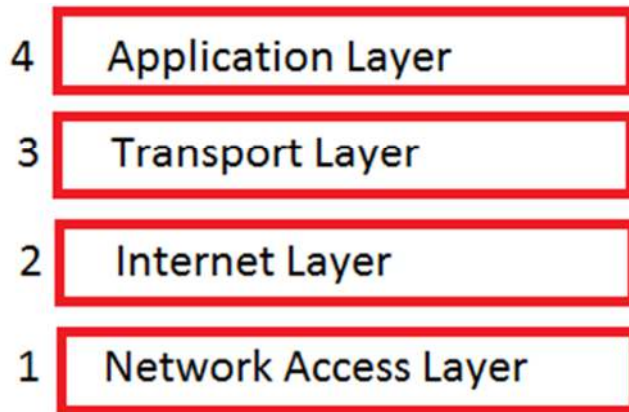
- Full-Duplex – simultaneous two-way communication
- Half-Duplex – two-way communication, but not simultaneous
- Simplex – one-way communication

6. Presentation layer:- The 6th layer of OSI model is called presentation layer. This layer is responsible for translation, encryption and compression. The protocol work under this later is SSL.

7. Application layer:-This is the final layer of OSI model. This layer provides user interface, and supports services such as mail services, file transfer, browsing internet, remote desktop connection etc. The protocols that communicate with the application layers are: SMTP, HTTP, POP3, IMAP, DHCP, and FTP.

The TCP/IP Protocol Suite

TCP/IP model consists of four layer::



1. Network Access layer:-

- This layer is the lowest layer of the TCP/IP model.
- The Network Access Layer describes how data is sent across the network.
- This layer is a combination of the data link layer and the physical layer defined in the OSI model.
- This layer is responsible for the transmission of data between two devices in the same network.
- The function of this layer is to encapsulate the IP datagram transmitted by the network into frames and map the IP address to the physical address.
- The protocols used by this layer are:- Ethernet, FDDI (Fiber Distributed Data Interface), x.25.

2. Internet Layer:-

- The data is packaged in the form of IP datagrams, this datagram contains the source and destination IP addresses so that the data can be easily sent and received.
- It is also called network layer.

The following protocols are used by this layer:-

IP protocol – Its full name is internet protocol and its main function is to deliver packets from source to destination. It has two versions IPv4 and IPv6.

ARP – Its full name is Address Resolution Protocol. Its task is to find the physical address from the IP address.

ICMP – Its full name is Internet Control Message Protocol. Its task is to inform the host about the problems coming in the network.

3. Transport Layer:-

This layer is responsible for the transmission of data,
It is also responsible for the reliability, flow control and correction of the data.

Two main protocols work in this layer:-

1. Transmission control protocol (TCP)
2. User Datagram Protocol (UDP)

4. Application Layer:-

- This layer is concerned with providing network service to the applications.
- This layer provides communication to the user; For example: - Through web browser, e-mail, and other applications.
- The application layer sends data to the transport layer and receives data from it.
- This layer allows the user to interact with the application.

The following are the protocols used in the application layer:-

HTTP and HTTPS:- It is used to access data on internet and it is also used to transfer text, audio and videos.

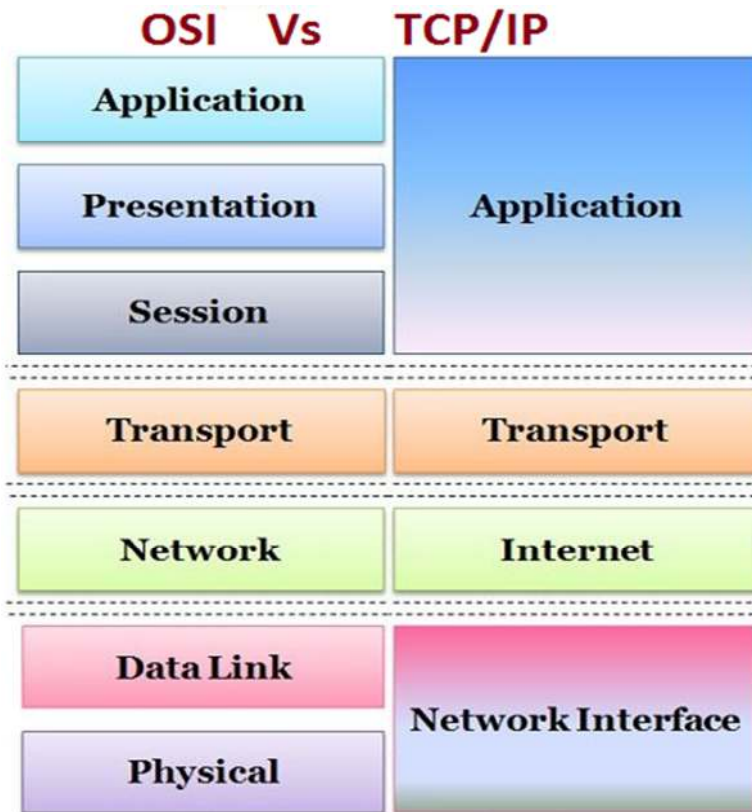
SNMP – Its full name is simple network management protocol. It is a framework used to manage devices on the Internet.

SMTP – Its full name is simple mail transfer protocol. It is used to send data from one e-mail to another e-mail address.

DNS – Its full name is domain name system. It is used to map the IP address.

SSH – Its full name is Secure Shell. It is used for encryption.

Comparison between OSI model and TCP/IP:-



OSI Layer	TCP/IP Layer
The full form of OSI is Open Systems Interconnection	The full form of TCP/IP is Transmission Control Protocol/ Internet Protocol
OSI model has been developed by ISO (International Standard Organization).	It was developed by ARPANET (Advanced Research Project Agency Network).
The OSI model is a conceptual model that characterizes and standardizes the communication function of a computing system.	TCP/IP is most widely used communication protocols, TCP/IP prepares and forward data packets over a network.
In the OSI model, the transport layer provides a guarantee for the delivery of the packets.	The transport layer does not provide the guarantee for the delivery of packets. But still, we can say that it is a reliable model.
This model is based on a vertical approach.	This model is based on a horizontal approach.
Protocols in the OSI model are hidden and can be easily replaced when the technology	In this model, the protocol cannot be easily replaced.

changes.	
Network Layer is both Connection Oriented and Connection less.	Network Layer is Connection less.
It is protocol dependent	It is protocol independent
It has 7 layers	It has 4 layers

Unit-1: Assignment

1. Explain the seven layers of the OSI model in detail with suitable diagrams.
2. Explain the TCP/IP model and describe all its layers in detail.
3. Compare the OSI model and TCP/IP model with a suitable diagram.
4. Explain different types of network topology with diagrams and advantages/disadvantages.